

Human Settlement on Another Celestial Body and
The Basic ways to Achieve It

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Abstract

Engineers start to move their focus onto settling other extraterrestrial planets as human technologies exponentially progress, and the earth soon becomes unable to support human life. As the plan of settlement is based on a completely new planet, many points must be seriously considered to ensure that human life can truly survive in the new environment and must pave the way for the future construction of society and cities. Environmental conditions which include both physics and biology affect survival, as humans can rarely survive in the high altitude and the pressure varies according to different densities of air. Energy support, which directly relates to the development of the whole world, must be considered due to different environmental conditions, and the way of generating energies may vary. To sustain the residential operations and mega cities, foundational infrastructures are critical because they ensure its safety of use and the ability to withstand extreme environments. Although settlement on an infinite celestial body is an unknown challenge, we can still use our own knowledge to overcome it.

Key words: extraterrestrial planet, air conditions, energy, and infrastructures

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Movies like *The Martian* have been featured all throughout the United States. The concept of human life on extraterrestrial planets has been featured throughout the world as our own planet Earth starts to die. The most promising solution to this is to build a new city. However, this won't be an easy feat, considering the atmospheric conditions, source of energy, and the construction of facilities.

Atmospheric conditions must be taken into mind when building a sustainable city.. This includes the pressure, radiation, and gravity of this new world. A human can only withstand 100 atm of pressure. A lower pressure would normally result in less oxygen. Inside buildings, a system built to stabilize pressured conditions can be used. As mentioned by Larson (2002), in airplanes, the system needs to take in air and compress it. However, compressing the air will cause the temperature to rise, so there must be something to cool the air. After this process, air is released, and stabilizes internal pressure. As mentioned in the article of Thermodynamic processes in closed systems - law of Amontons for ideal gas (2020), if the volume stays the same, pressure and temperature are directly proportional. Therefore, as pressure increases, so does the temperature. A pressure humans are able to withstand must be reached. To properly design a city, the gravitational force must also be taken into consideration. As gravity increases, the amount of force exerted on objects will increase. Thus, this calls for more durable materials when building. Although many may not think of it, increased radiation exposure will cause hazards such as terminal diseases. To prevent this, advanced shielding materials can be used. Another effective way to avert exposure is building a city underground.

To live in a modern world with comfort, there must be a source of energy. Without the use of energy, there is no way to advance as a society. The first priority is to transition to renewable energy sources. Research by Harvey & Dhar (2022) supports that an efficient way to extract energy is using solar panels. When sunlight, made up of photons, hits the silicon in a solar panel, it makes the electrons excited, causing them to release energy when moving from the orbitals. Not only does it prove a reliable energy source, solar energy is easily the best, being a renewable energy. The sun has more than enough energy to satisfy even the most luxurious cities. However, an alternative is energy extracted from nuclear power using atoms of uranium which are forced to split therefore creating heat energy. According to the article How does a nuclear plant make electricity? (2020) in NRC, the heat coming out from the nuclear reaction can make steam from boiling water, and that steam runs through a machine called turbine, which acts something like a highly- sophisticated windmill. As the steam causes part of the turbine to spin, that spinning part of the machine can be connected to a generator to produce electricity.

Undoubtedly, the buildings and skyscrapers hold great value in a city. Preparing for natural disasters, climate, and environment are the major aspects to consider while designing structures to build. If an environment is prone to earthquakes, there must be base isolators, used to prevent damage. According to the Christian Science Monitor (2012), a device called a fuse may be used to protect buildings from seismic activity. During the earthquake, it will absorb the impact, leaving the buildings unharmed. Afterwards, the damaged fuses will need to be repaired.

The construction of reliable water and electricity systems is paramount for the establishment of a sustainable new city on an extraterrestrial planet. A crucial element involves

designing a water distribution network to ensure a consistent and stable supply of water throughout the city. Underground pipelines are a viable option for transporting water resources, but they must be meticulously planned and constructed to accommodate the unique geological and topographical features of the extraterrestrial environment. The choice of materials for these pipelines is critical, with a focus on selecting materials that are resistant to corrosion and capable of withstanding high pressures to ensure the integrity and longevity of the system. Similarly, when considering the electrical infrastructure, it is essential to choose materials that can withstand the challenges of the extraterrestrial environment. Given the potential for high levels of radiation, it is imperative to use radiation-resistant materials in the construction of electrical components such as cables and circuits. These materials should be durable and adaptable to extreme temperature and pressure conditions to ensure the reliability and stability of the electrical system over time.

In the end of the Martian, the man went back to planet Earth, but building a completely new city will be our home. As engineers, we want to protect our home and families. As we build a city, designing challenges must be accounted for such as the quality of life, including the natural environment, energy extraction, and construction design. As we expand our home, we must also expand our knowledge and horizons.

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